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GRASP: Graph Reasoning via Agentic Solving and Probing of LLMs

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📄 Abstract

Integrating graph knowledge into Large Language Models (LLMs) via passive representation faces critical bottlenecks: limited context windows, unreliable numerical computation, and structural hallucinations. To solve this, we propose **GRASP** (Graph Reasoning via Agentic Solving and Probing), shifting the paradigm from passive ingestion to proactive agentic exploration. By interleaving Neighbor Retrieval for on-demand probing with Code Interpreter as a deterministic solver, GRASP enables LLMs to autonomously navigate and compute over complex topologies. We employ a staged reinforcement learning strategy (GRPO) that transitions from visible tuning to a structure-blind environment, forcing the agent to develop genuine topological awareness. Evaluated on multi-domain graph reasoning benchmarks, our 4B model achieves a 53.06\% average performance boost, surpassing SOTA baselines like DeepSeek-V3.2 and successfully generalizing to unseen tasks, with high potential for tackling sampling on million-node graphs and solving Hard-level LeetCode graph problems.

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